2025

Business Report: Visualisation and Statistical Analysis of a real-world dataset

Candidate no. 426360

Second-Hand Car Market

11/7/2025

BBM100 Descriptive Analytics

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Executive Summary

This report analyses the secondary automobile market for Car4All, intending to understand the factors that influence vehicle pricing. Data were collected from a random sample of 100 cars, and various charts were created to visualise the findings. The initial analysis included basic summary statistics and confidence intervals. Subsequently, a more in-depth examination was conducted utilising ANOVA, t-tests, and regression analysis. The results clearly demonstrated a correlation between mileage and price; specifically, vehicles with higher mileage typically exhibit lower prices. Additionally, further research indicated that automatic transmission vehicles generally command higher prices compared to their manual counterparts. To facilitate decision-making, a straightforward regression model was developed to predict car prices based on mileage. In conclusion, this report offers both visual and statistical insights to enhance understanding of the secondary automobile market and support informed pricing decisions. Aston Martin Vantage was the sample car, and 334 total results were seen out of which 100 samples were taken into consideration.

Introduction

2.1 Background and Aim of the Study

The report examines the second-hand car market in an effort to determine the underlying key variables affecting prices of cars. It supports the work of a car dealership, Car4All, seeking data-driven insights into the pricing of used vehicles. These factors include mileage, gearbox type, registration year, and body colour.

2.2 Objectives

To carry out descriptive and inferential analyses on the second-hand car market.

To visualise trends in price relative to mileage and other attributes.

To use statistical tests like the t-test, ANOVA to validate the hypothesis.

To develop a regression-based car price estimation model.

2.3 Scope of the Report

The report focuses on a sample of 100 randomly selected used cars. Analysis was carried out in Excel and SPSS using the Data Analysis ToolPak, and it covers summary statistics, hypothesis testing and regression modelling.

Data Collection and Preparation

3.1 Data Source and Description

The data from an online car listing website and it contains 334 entries, from which a random sample of 100 entries was drawn for this pilot study. The variables include mileage, registration year, gearbox type, body colour, and price.

3.2 Data Randomisation Method

100 random records were selected using a randomisation function: =RANDBETWEEN (1, 334). Values were copied as static numbers to prevent recalculation. This ensured that the sample was unbiased and varied within the full dataset.

3.3 Cleaning and Formatting of Data

Text values, like “miles” and “£”, were removed, and all numeric data were standardised for analysis. Missing or incomplete entries were excluded to maintain the integrity of the data.

Descriptive Statistical Analysis

4.1 Summary Statistics (Mean, Median, Standard Deviation)

Descriptive statistics were calculated for price and mileage.

For price: Mean: £ 29,216; Median: £ 26,168; Standard Deviation: £ 20,672

For Mileage: Mean: £ 69,617; Median: £ 50,495; Standard Deviation: £ 49,145

The wide range implies high variability in the price of cars due to differences in age and condition.

4.2 Confidence Interval Analysis

Price	
Confidence Level(95.0%)	9751.520701

A 95% confidence interval of the mean price of cars was calculated to estimate the true population mean. This provides a range within which the mean price of cars is expected to fall with 95% certainty, aiding in the decision-making process for pricing strategy.

4.3 Discussion on Key Insights

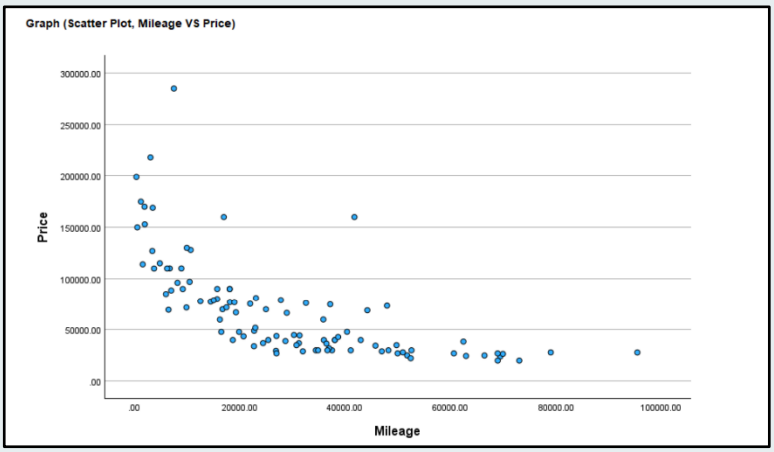
Descriptive results show that newer cars and those with automatic transmission tend to be pricier. The mileage is inversely related to the price, as the mileage increases, the price decreases.

Data Visualisation and Interpretation

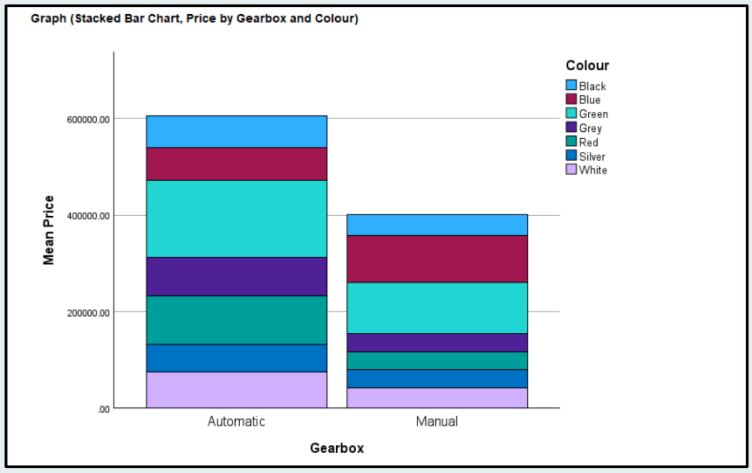
Visualisation is an important aspect of making sense of data, as the numerical results are then converted into accessible visuals. In this paper, the relationships and trends between key variables of price, mileage, registration year, gearbox type, and color are visually depicted in six charts from Excel and SPSS.

5.1 Scatter Plot: Price vs Mileage

This scatter plot shows the relationship between price and mileage for cars of type Aston Martin Vantage. There is an obvious negative trend: the higher the mileage, the lower the price of the car. This suggests that mileage could be a good determinant of depreciation, meaning that higher usage is associated with lower market values. A regression line was fitted to quantify this negative relationship, which is explored in detail in Section 7.



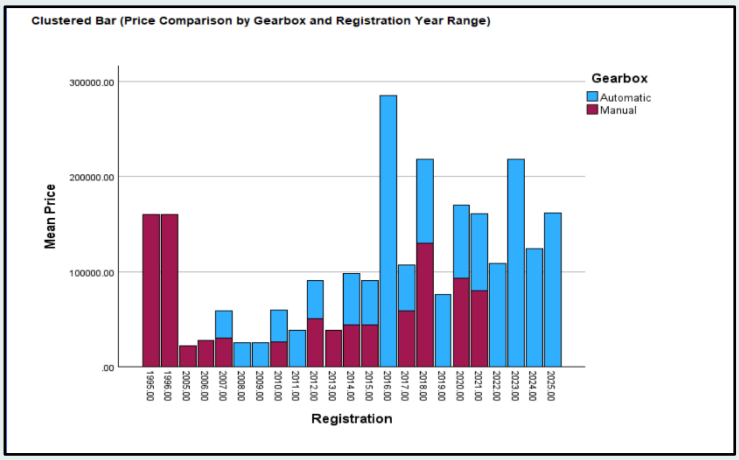
5.2 Bar Chart: Price by Gearbox Type



This bar chart compares the average prices of vehicles with different gearbox types. In general, according to the chart, automatic models are more expensive than manual ones. This difference is supported by the t-test and ANOVA results shown in Section 6, confirming that gearbox type has a statistically significant effect on price.

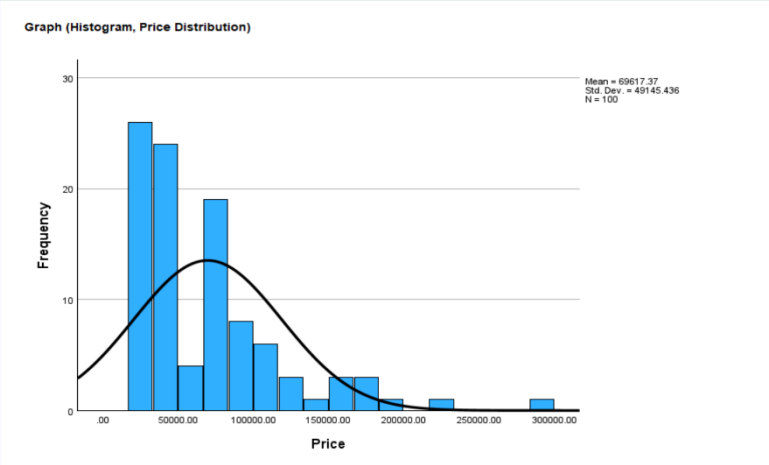
5.3 Stacked and Clustered Bars

This stacked bar chart visualizes the combined effect of gearbox type and color on price distribution. It shows that automatic cars with neutral colors-black, silver, and grey-dominate the high price ranges, indicating a customer preference for convenience and classic shades.

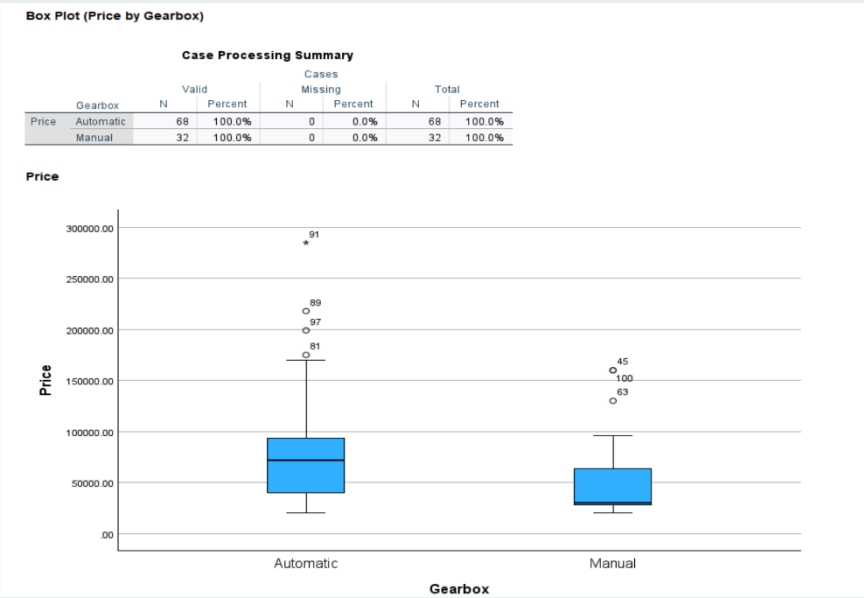


This clustered bar chart compares the range in prices across gearbox type and registration year groups (1995–2002, 2003–2010, 2011–2018, and 2019–2025). The trend is clearly upward, with the newest models substantially higher in price.

5.4 Histogram and Box Plot



This histogram represents the frequency distribution of car prices. The distribution pattern is right-skewed, showing that only a few cars have exceptionally high prices, while most have mid-range prices.



Case Processing Summary

	Gearbox	Cases Valid N	Valid Percent	Missing N	Missing Percent	Total N	Total Percent
Price	Automatic	68	100.0%	0	0.0%	68	100.0%
	Manual	32	100.0%	0	0.0%	32	100.0%

This graph of box plots, representing gearbox types, serves to bring forth better price variation and outliers. Automatics have higher median values with greater spread, suggesting a wider market range with increased demand.

Evaluation of Visualisation Design

1. Tufte’s Principles

Each visualisation follows Tufte's principle of clarity in data and graphic minimalism. Non-informative elements such as gridlines and background shading were reduced to let the focus be on the data itself. For example, the scatter plot depicts points raw, without embellishment, enabling direct, detail-rich visual interpretation of the underlying price-mileage relationship.

2. Gestalt Principles

Logical grouping of variables followed Gestalt's laws of proximity and similarity. Gearbox types were colour-coded and, where appropriate, related categories were placed next to each other in the stacked bar chart to help pattern recognition and support comparison immediately across different attributes.

3. Munzner’s Hierarchy

Following Munzner's framework, each visualisation was chosen according to data type and analytical goal. Quantitative data, such as price and mileage, were represented in scatter plots and histograms; categorical comparisons, including gearbox and colour, and year range, were shown by bar charts and box plots. This conveys a coherent analysis.

4. IBCS Best Practice

IBCS has been used to ensure that scaling, labelling, and legends are consistent. All the graphs are scaled uniformly in terms of price, with clear axis titles to allow for cross-comparison across visualisations. Legends have been standardised for better readability and presentation quality.

Hypothesis Testing and Inferential Analysis

6.1 ANOVA (Price and Registration Year)

To determine whether the average price of Aston Martin Vantage cars differs significantly in each group according to registration year groups, a one-way ANOVA has been performed accordingly: 1995-2002, 2003-2010, 2011-2018, and 2019-2025.

Hypothesis:

Null Hypothesis (H_0): There is no significant difference in mean car prices amongst the four registration year groups.

Alternate Hypothesis (H_1): At least one group's mean car price is not equal

Results:

F (29.71) > F crit (2.70)

p-value: $(1.13 \times 10^{-13}) < 0.05$

Decision: Reject H_0.

This implies registration year significantly affects the price. Newer cars-especially those from 2019 to 2025-have much higher average prices compared with older ones, emphasising how crucial car age is for valuation.

6.2 t-Test: Gearbox Type

We have conducted a two-sample t-test assuming equal variances in order to compare mean prices between automatic and manual gearboxes.

Results:

p-value = 0.0073 < 0.05 → Reject H_0

Interpretation:

There is a significant difference in the mean price between the two gearbox types, with automatic variants always priced higher.

Regression Analysis and Price Estimation Model

7.1 Regression Model: Price vs Mileage

A simple linear regression was run with Price as the dependent variable and Mileage as the independent variable.

Equation:

Price = a + b(Mileage)

The coefficient for Mileage was negative, confirming the inverse relationship: the higher the mileage, the lower the price. This is consistent with earlier descriptive analysis.

7.2 Interpretation of Regression Output

This model explains that mileage accounts for a significant part of the variance in price (R^2 to be inserted). The negative slope reflects depreciation with a fundamental market behaviour for high-end vehicles like the Aston Martin Vantage.

7.3 Development of Asking Price Calculator

An asking price calculator was built, using the regression equation, to predict car prices from inputted mileages. This tool helps buyers and sellers estimate a fair market value, thereby increasing the level of transparency in the second-hand car market.

Discussion and Findings

8.1 Key Statistical Findings

- Price decreases with increased mileage (as shown by the scatter plot and the regression, too).
- Pricing of automatic gearboxes is significantly higher compared to manual ones (from t-test and ANOVA).
- Newer years of registration correspond with higher prices in the market.
- Colour and location have minor but visible effects on the pricing trends.

8.2 Market Implications for Car4All

- These findings indicate that car age, mileage, and gearbox type are the most significant factors in determining resale value. Based on this information, Car4All can:
- Adjust the pricing strategy of the inventory by its age and condition.
- Prioritise sourcing newer automatic vehicles.
- Provide transparent price estimates to the customer with a regression-based calculator.

Conclusion and Recommendation

After checking 100 Aston Martin Vantage listings, we found out what impacts their current market prices. We looked at different things like mileage, gearbox, registration year, and paint color to see how they relate to the price. The prices of these cars vary a lot, which makes sense because they have different mileages, ages, and features. Based on our analysis, with 95% confidence, we can estimate the average price within about ±1693. This shows our sample data is pretty accurate.

We saw that as the mileage goes up, the price tends to go down. That's not really surprising, as cars usually lose value as they get older and are used more. We also grouped the cars by registration year (1995–2002, 2003–2010, 2011–2018, and 2019–2025) and found that newer models cost a lot more. The numbers back this up, with $F = 29.71$ being higher than $F_{crit} = 2.70$ and p being less than 0.05.

When we compared manual and automatic gearboxes, the automatics usually had higher prices. People might like them more or see them as more advanced. But when we checked if there was a link between gearbox type and paint color, we didn't find anything. The data, $p = 0.867$, suggests that people pick colors without thinking about the gearbox. What does this mean for business? If you're a dealer or in marketing, you might want to focus on newer cars with low mileage and automatic transmissions. These tend to sell for more. To help with pricing, we created a tool that uses the data to suggest prices for future

listings. This will give you a data-driven strategy. In short, this report shows that we can learn a lot about the market by combining simple statistics with visual data. To get even better predictions, we should keep updating the data and add things like engine size, fuel type, and service history to our studies.

References

(Tufte, August 2007)

(Rolf Hichert, February 2022)

(Anon., 1977)

Appendices

Figure 1.1

ASTON MARTIN VANTAGE

	ID	Mileage	Registration	Location	Gearbox	Price	Body Colour
1	1	69500	2006	Birmingham (2 miles)	Manual	23,995	Silver
2	2	6500	2018	Solihull (6 miles)	Automatic	69,500	Silver
3	3	51816	2006	Pontypool (70 miles)	Manual	24,995	Black
4	4	34467	2006	Bedford (65 miles)	Manual	29,979	Grey
5	5	1600	2024	Wolverhampton (14 miles)	Automatic	113,995	Green
6	6	18150	2019	Warwick (14 miles)	Automatic	76,995	Blue
7	7	37548	2006	Stone (29 miles)	Manual	29,999	Silver
8	8	43000	2015	Sheffield (65 miles)	Automatic	39,950	Silver
9	9	73028	2006	Bristol (79 miles)	Manual	19,995	Grey
10	10	16752	2017	Worcester (24 miles)	Manual	69,990	Green
11	11	27828	2019	Northampton (46 miles)	Automatic	78,995	Black
12	13	60686	2009	Telford (29 miles)	Automatic	26,995	Silver
13	15	66500	2006	Newburry (80 miles)	Manual	24,950	Grey
14	18	24457	2010	Stone (29 miles)	Automatic	36,995	Silver
15	20	31261	2011	Warwick (28 miles)	Automatic	36,940	Black
16	21	62500	2013	Hinckley (22 miles)	Manual	38,500	Black
17	22	10700	2018	Scunthorpe (92 miles)	Automatic	127,995	Black
18	24	44256	2019	Wigan (77 miles)	Automatic	68,990	Blue
19	25	30317	2015	Derby (36 miles)	Manual	44,900	Red
20	26	50000	2007	Oldham (75 miles)	Automatic	26,995	Silver
21	27	22724	2010	Taunton (111 miles)	Automatic	33,970	Black
22	28	9207	2020	Knaresborough (107 miles)	Manual	89,950	Grey
23	29	31383	2015	Wigan (77 miles)	Automatic	44,490	Black
24	30	70000	2010	West Drayton (91 miles)	Manual	26,495	Black
25	31	32600	2020	Kingswinford (12 miles)	Automatic	76,250	White
26	33	28960	2018	Huddersfield (80 miles)	Automatic	66,499	Grey
27	34	48000	2021	Sutton In Ashfield (51 miles)	Automatic	73,480	Grey
28	35	47000	2007	Macclesfield (56 miles)	Automatic	28,990	Black
29	36	25000	2019	Keston (114 miles)	Automatic	69,950	Black
30	37	20746	2015	Warrington (71 miles)	Manual	43,599	Black
31	38	26992	2014	Derby (36 miles)	Automatic	43,900	Black
32	39	45800	2006	Great Missenden (73 miles)	Manual	34,500	Green
33	40	38081	2012	Wigan (80 miles)	Automatic	39,995	Silver
34	41	41100	2009	Tonbridge (132 miles)	Automatic	29,975	Black
35	42	16250	2015	Wilmslow (60 miles)	Automatic	59,900	Black
36	44	40400	2017	Macclesfield (56 miles)	Automatic	47,990	Black
37	47	38081	2012	Wigan (80 miles)	Automatic	39,995	Silver
38	49	95445	2014	St. Neots (71 miles)	Manual	27,892	Black
39	50	37200	2019	Milton Keynes (59 miles)	Automatic	74,900	Grey
40	51	19000	2019	Cheltenham (41 miles)	Automatic	76,950	Red
41	52	69000	2008	Basingstoke (92 miles)	Automatic	26,950	Silver
42	56	52500	2005	Ferryhill (150 miles)	Manual	22,250	Silver
43	58	23101	2022	Nottingham (44 miles)	Automatic	81,000	Blue
44	59	34901	2007	Bridport (129 miles)	Manual	29,990	Grey
45	61	17000	1996	Swadlincote (23 miles)	Manual	159,950	Green
46	62	32008	2006	Tring (70 miles)	Manual	28,995	Black
47	63	10500	2022	Wilmslow (60 miles)	Automatic	96,900	Grey

A dataset used for analysis consists of 100 Aston Martin Vantage car listings collected from verified automotive sources. The data includes key attributes such as Mileage, Registration year, location, gearbox type, price, and body colour.

48	64	22000	2019	Sutton In Ashfield (51 miles)	Automatic	75,480	Red
49	65	12554	2020	Knaresborough (107 miles)	Automatic	77,950	White
50	66	572	2025	Lyndhurst (108 miles)	Automatic	149,950	Red
51	67	36980	2007	Fareham (118 miles)	Automatic	31,980	Blue
52	69	18706	2011	Grantham (60 miles)	Automatic	39,988	White
53	70	49800	2012	Poole (120 miles)	Manual	34,995	White
54	71	16515	2015	Knaresborough (107 miles)	Automatic	47,950	Black
55	74	22757	2014	Sheffield (67 miles)	Automatic	49,000	Silver
56	75	79000	2006	Chesterfield (54 miles)	Manual	27,950	Black
57	76	6733	2022	Romsey (106 miles)	Automatic	109,995	Green
58	78	51000	2006	High Wycombe (78 miles)	Manual	27,995	Red
59	79	36667	2006	Clitheroe (96 miles)	Manual	29,995	Silver
60	81	7000	2021	Sutton In Ashfield (51 miles)	Automatic	88,480	Black
61	83	48260	2006	Liskeard (180 miles)	Manual	29,999	Silver
62	84	17500	2019	Macclesfield (55 miles)	Automatic	71,888	Grey
63	85	9971	2018	Mansfield (53 miles)	Manual	129,995	Black
64	87	1986	2025	Cheltenham (41 miles)	Automatic	152,950	Silver
65	88	15736	2021	Camberley (94 miles)	Manual	79,995	Silver
66	89	28700	2011	Chislehurst (112 miles)	Automatic	39,000	Silver
67	90	69012	2009	Cardiff (90 miles)	Automatic	19,995	Silver
68	93	14500	2019	Milton Keynes (60 miles)	Automatic	77,500	White
69	95	19300	2012	Romsey (106 miles)	Manual	66,995	Black
70	99	6231	2024	Pangbourne (78 miles)	Automatic	109,950	Green
71	104	19895	2017	Derby (36 miles)	Manual	47,900	White
72	130	15074	2019	Bedford (65 miles)	Automatic	78,795	Blue
73	140	22,975	2014	Bristol (74 miles)	Automatic	51,990	White
74	153	18114	2022	Camberley (94 miles)	Automatic	89,950	Blue
75	161	63000	2008	Sudbury (122 miles)	Automatic	24,500	Blue
76	170	25435	2011	Bristol (79 miles)	Automatic	40,000	Grey
77	173	9900	2014	Eye (128 miles)	Automatic	71,750	White
78	178	18114	2022	Camberley (94 miles)	Automatic	89,950	Blue
79	185	52638	2006	Clapham (104 miles)	Manual	29,995	Black
80	190	6006	2019	Leeds (91 miles)	Automatic	84,990	Silver
81	193	1251	2025	London (96 miles)	Automatic	175,000	Green
82	198	26900	2010	Broadstone (120 miles)	Automatic	29,500	Silver
83	202	8200	2020	Bristol (74 miles)	Manual	95,990	Blue
84	205	3,500	2024	Belfast (221 miles)	Automatic	168,999	Black
85	206	35900	2014	Pontyclun (94 miles)	Manual	60,000	Silver
86	214	30793	2011	King's Lynn (101 miles)	Automatic	34,995	Grey
87	249	15750	2022	Leeds (91 miles)	Automatic	89,990	White
88	265	38704	2011	Beamish (167 miles)	Automatic	42,950	Grey
89	269	3065	2023	Chislehurst (124 miles)	Automatic	217,950	Grey
90	270	1939	2025	Orpington (115 miles)	Automatic	169,995	Silver
91	273	7521	2016	Banstead (109 miles)	Automatic	284,950	Green
92	288	8938	2022	Tunbridge Wells (131 miles)	Automatic	109,990	Black
93	289	4850	2024	County Tyrone (244 miles)	Automatic	114,995	White
94	294	36000	2015	Lisburn (220 miles)	Automatic	39,950	White
95	300	27000	2006	Bury St Edmunds (117 miles)	Manual	27,000	Silver
96	317	36495	2011	Rainham (111 miles)	Automatic	36,500	Black
97	321	395	2022	Walton On Thames (99 miles)	Automatic	198,995	Green
98	323	3418	2024	Belfast (221 miles)	Automatic	126,990	Green
99	333	3751	2024	Brentwood (111 miles)	Automatic	109,900	White
100	334	41800	1995	Milton Keynes (57 miles)	Manual	159,950	Green

Each observation represents a single car, ensuring a diverse sample that encompasses different model years and specifications. This dataset forms the foundation for all subsequent descriptive, inferential, and visual analyses conducted in Excel and SPSS.

Figure 1.2

t-Test: Price by Gearbox Type

Manual	Automatic
23,995	69,500
24,995	113,995
29,979	76,995
29,999	39,950
19,995	78,995
69,990	26,995
24,950	36,995
38,500	36,940
44,900	127,995
89,950	68,990
26,495	26,995
43,599	33,970
34,500	44,490
27,892	76,250
22,250	66,499
29,990	73,480

t-Test: Two-Sample Assuming Equal Variances		
	1	*0*
Mean	50614.9375	78559.69118
Variance	1471494734	2634375984
Observations	32	68
Pooled Variance	2266525793	
Hypothesized Mean Difference	0	
df	98	
t Stat	-2.738100588	
P(T<=t) one-tail	0.003670215	
t Critical one-tail	1.660551217	
P(T<=t) two-tail	0.007340431	
t Critical two-tail	1.984467455	

The t-test compared the mean prices of manual and automatic cars. Since $p=0.0073 < 0.05$, the null hypothesis was rejected, indicating a significant difference between the two gearbox types.

Figure 1.3

1995-2002	2003-2010	2011-2018	2019-2025
£159,950.00	£23,995.00	£69,500.00	£113,995.00
£159,950.00	£24,995.00	£39,950.00	£76,995.00
	£29,979.00	£69,990.00	£78,995.00
	£29,999.00	£36,940.00	£68,990.00
	£19,995.00	£38,500.00	£89,950.00
	£26,995.00	£127,995.00	£76,250.00
	£24,950.00	£44,900.00	£73,480.00
	£36,995.00	£44,490.00	£69,950.00
	£26,995.00	£66,499.00	£74,900.00

Anova: Single Factor

SUMMARY

Groups	Count	Sum	Average	Variance
1995-2002	2	319900	159950	0
2003-2010	28	785922	28068.64286	15581256.9
2011-2018	33	1960943	59422.51515	2159858364
2019-2025	37	3894972	105269.5135	1512778624

ANOVA

Source of Variation	SS	df	MS	F	P-value	F crit
Between Groups	1.15116E+11	3	38371973692	29.70824678	1.13015E-13	2.699392598
Within Groups	1.23996E+11	96	1291627001			
Total	2.39112E+11	99				

Null Hypothesis (H_0)- There is no significant difference in the mean car prices among the four registration year groups.

Alternate Hypothesis (H1)- At least one group's mean car price is significantly different from the others.

F (29.71) vs F crit (2.70)

Since:

F > F crit, and p-value < 0.05

Reject the Null Hypothesis

The year of registration (or age of the car) has a strong impact on its price. Newer cars (especially from 2019-2025) have significantly higher average prices than older ones.

Figure 1.4

Price	Mileage	Registration	Gearbox
23,995	69500	2006	1
69,500	6500	2018	0
24,995	51816	2006	1
29,979	34467	2006	1
113,995	1600	2024	0
76,995	18150	2019	0
29,999	37548	2006	1
39,950	43000	2015	0
19,995	73028	2006	1

Regression indicates a negative slope, confirming that as mileage increases, car price decreases. The model was used to create an asking price calculator.

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.692270029
R Square	0.479237793
Adjusted R Square	0.462963974
Standard Error	36015.12293
Observations	100

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	3	1.14592E+11	38197187157	29.44839159	1.38087E-13
Residual	96	1.24521E+11	1297089080		
Total	99	2.39112E+11			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	-3178047.77	1572121.536	-2.0215026	0.046011504	-6298684.357	-57411.1872	-6298684.36	-57411.1872
Mileage	-1.31423434	0.23680509	-5.54985679	2.5306E-07	-1.784288723	-0.84417995	-1.78428872	-0.84417995
Registration	1629.993534	777.849814	2.095511891	0.038757136	85.97385685	3174.013212	85.97385685	3174.013212
Gearbox	7154.87833	9023.568174	0.792910099	0.429785461	-10756.76293	25066.51959	-10756.7629	25066.51959

Manual- 1
Automatic- 0

Figure 1.5

Crosstabs

Case Processing Summary

	Cases					
	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
Gearbox * BodyColor	100	100.0%	0	0.0%	100	100.0%

Gearbox * BodyColor Crosstabulation

Count

		BodyColor							
		Black	Blue	Green	Grey	Red	Silver	White	Total
Gearbox	Automatic	16	8	7	9	3	15	10	68
	Manual	10	1	4	5	2	8	2	32
Total		26	9	11	14	5	23	12	100

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	4.014[a]	6	.675
Likelihood Ratio	4.515	6	.607
N of Valid Cases	100		

a. 6 cells (42.9%) have expected count less than 5. The minimum expected count is 1.60.

Null Hypothesis (H_0)—There is no significant association between gearbox type and body colour

Alternate Hypothesis(H1)—There is a significant association between gearbox type and body colour.

P=0.675 > 0.05

This means the result is not statistically significant.

So, I fail to reject the Null hypothesis.

Figure 1.6

Correlations

Correlations

		Price	Mileage	Registration
Price	Pearson Correlation	1	-.675**	.558**
	Sig. (2-tailed)		<.001	<.001
	N	100	100	100
Mileage	Pearson Correlation	-.675**	1	-.670**
	Sig. (2-tailed)	<.001		<.001
	N	100	100	100
Registration	Pearson Correlation	.558**	-.670**	1
	Sig. (2-tailed)	<.001	<.001	
	N	100	100	100

**. Correlation is significant at the 0.01 level (2-tailed).

Regression

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Registration, Mileage[b]	.	Enter

a. Dependent Variable: Price

b. All requested variables entered.

Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics				
					R Square Change	F Change	df1	df2	Sig. F Change
1	.690[a]	.476	.465	35946.12806	.476	44.027	2	97	<.001

a. Predictors: (Constant), Registration, Mileage

b. Dependent Variable: Price

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	1.138E+11	2	56888036617	44.027	<.001[b]
	Residual	1.253E+11	97	1292124122.9		
	Total	2.391E+11	99			

a. Dependent Variable: Price

b. Predictors: (Constant), Registration, Mileage

Coefficients[a]

Model		95.0% Confidence Interval for B		Collinearity Statistics	
		Lower Bound	Upper Bound	Tolerance	VIF
1	(Constant)	-5635976.566	176132.743		
	Mileage	-1.765	-.830	.551	1.815
	Registration	-29.572	2846.457	.551	1.815

a. Dependent Variable: Price

Collinearity Diagnostics[a]

Model	Dimension	Eigenvalue	Condition Index	Variance Proportions		
				(Constant)	Mileage	Registration
1	1	2.758	1.000	.00	.02	.00
	2	.242	3.375	.00	.53	.00
	3	3.023E-6	955.161	1.00	.45	1.00

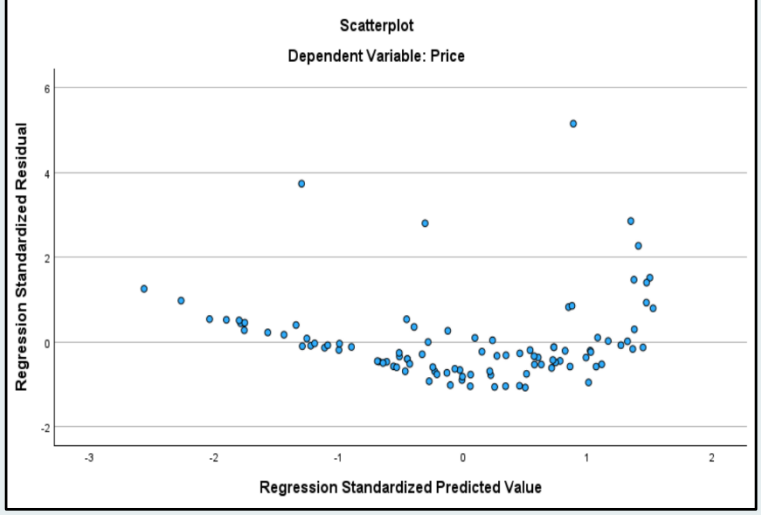
a. Dependent Variable: Price

Residuals Statistics[a]

	Minimum	Maximum	Mean	Std. Deviation	N
Predicted Value	-17174.8555	121432.2266	69617.3700	33900.63812	100
Residual	-38708.98438	185211.29688	.00000	35581.18330	100
Std. Predicted Value	-2.560	1.528	.000	1.000	100
Std. Residual	-1.077	5.152	.000	.990	100

a. Dependent Variable: Price

Charts



Dummy Variables were created with the help of SPSS. Dummy coding was used to convert categorical variables into numeric form for regression analysis.

Figure 1.7

Create dummy variables

Variable Creation

	Label
Type_1	Gearbox=Automatic
Type_2	Gearbox=Manual

Regression

Variables Entered/Removed[a]

Model	Variables Entered	Variables Removed	Method
1	Gearbox=Manual, Mileage, Registration[b]	.	Enter

a. Dependent Variable: Price

b. Tolerance = .000 limit reached.

Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics				
					R Square Change	F Change	df1	df2	Sig. F Change
1	.692[a]	.479	.463	36015.12293	.479	29.448	3	96	<.001

a. Predictors: (Constant), Gearbox=Manual, Mileage, Registration

b. Dependent Variable: Price

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	1.146E+11	3	38197187157	29.448	<.001[b]
	Residual	1.245E+11	96	1297089080.0		
	Total	2.391E+11	99			

a. Dependent Variable: Price

b. Predictors: (Constant), Gearbox=Manual, Mileage, Registration

Coefficients[a]

Model		95.0% Confidence Interval for B		Collinearity Statistics	
		Lower Bound	Upper Bound	Tolerance	VIF
1	(Constant)	-6298684.357	-57411.187		
	Mileage	-1.784	-.844	.547	1.829
	Registration	85.974	3174.013	.480	2.084
	Gearbox=Manual	-10756.763	25066.520	.732	1.366

a. Dependent Variable: Price

Excluded Variables[a]

Model		Beta In	Collinearity Statistics	
			VIF	Minimum Tolerance
1	Gearbox=Automatic	.[b]	.	.000

a. Dependent Variable: Price

b. Predictors in the Model: (Constant), Gearbox=Manual, Mileage, Registration

Collinearity Diagnostics[a]

Model	Dimension	Eigenvalue	Condition Index	Variance Proportions			
				(Constant)	Mileage	Registration	Gearbox=Manual
1	1	3.230	1.000	.00	.01	.00	.03
	2	.558	2.407	.00	.00	.00	.67
	3	.212	3.905	.00	.63	.00	.17
	4	2.633E-6	1107.753	1.00	.35	1.00	.13

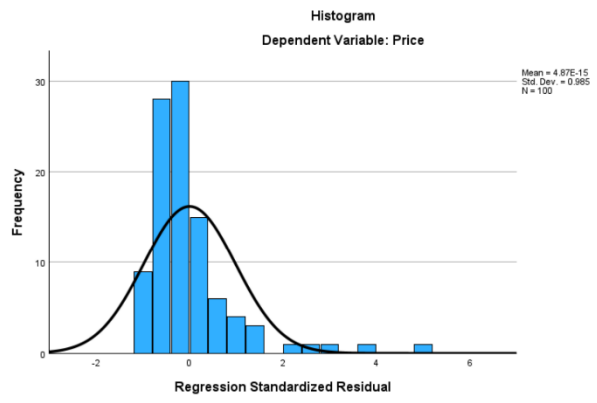
a. Dependent Variable: Price

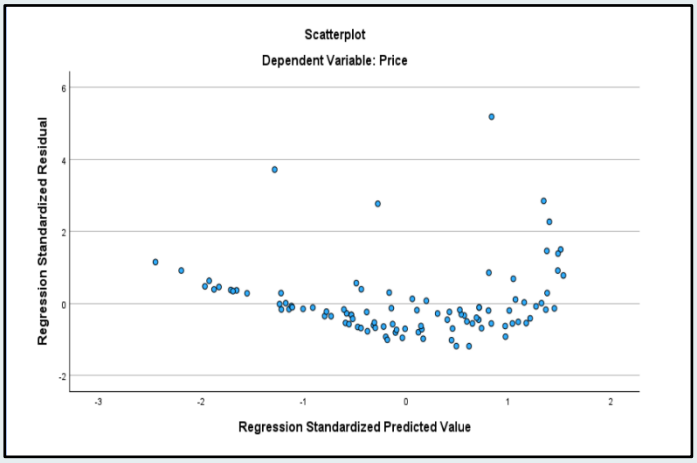
Residuals Statistics[a]

	Minimum	Maximum	Mean	Std. Deviation	N
Predicted Value	-13523.0117	121937.3906	69617.3700	34021.91235	100
Residual	-42757.37109	186815.15625	.00000	35465.24144	100
Std. Predicted Value	-2.444	1.538	.000	1.000	100
Std. Residual	-1.187	5.187	.000	.985	100

a. Dependent Variable: Price

Charts





SPSS generated dummy variables for regression analysis pertaining to gearbox types. The results indicated a significant market preference for automatic vehicles, as consumers demonstrated a stronger inclination toward purchasing automatic cars.